HDOC Day-75

13 November 2026 | C Programming - Enumeration

Question 1: Enumeration Values

Assign explicit values starting from 10 in an enumeration and print them. Input: Nil; Output: A=10, B=11, C=12.
Program:

```
#include <stdio.h>

enum Numbers { A = 10, B, C };

int main(void) {
    printf("A=%d, B=%d, C=%d\n", A, B, C);
    return 0;
}
```

Input: Nil
Output: A=10, B=11, C=12
Explanation: A is assigned 10 explicitly. The following enumeration constants automatically receive consecutive values, 11 and 12.

Question 2: Menu Operations Using enum and switch

Use enumeration to represent menu choices (ADD, SUBTRACT, MULTIPLY) and perform operations using switch. Sample input: ADD 10 20; Output: 30.
Program:

```
#include <stdio.h>
#include <string.h>

enum Operation { ADD, SUBTRACT, MULTIPLY };

int main(void) {
    char choice[20];
    int a, b;
    enum Operation op;

    if (scanf("%19s %d %d", choice, &a, &b) != 3) {
        printf("Invalid input\n");
        return 1;
    }

    if (strcmp(choice, "ADD") == 0)
        op = ADD;
    else if (strcmp(choice, "SUBTRACT") == 0)
        op = SUBTRACT;
    else if (strcmp(choice, "MULTIPLY") == 0)
        op = MULTIPLY;
    else {
        printf("Invalid choice\n");
        return 1;
    }

    switch (op) {
        case ADD:      printf("%d\n", a + b); break;
        case SUBTRACT: printf("%d\n", a - b); break;
        case MULTIPLY: printf("%d\n", a * b); break;
```

```
    }
    return 0;
}
```

Sample Input: ADD 10 20
Sample Output: 30
Explanation: The input word is converted into an enumeration value. The switch statement selects the matching arithmetic operation and prints its result.